

Department of Electrical and Electronic Engineering

Islamic University of Technology (IUT)

Name of the Project: SAP-1 Architecture-based 6-bit Computer Design and Implementation

Supervisor: Mr. Hayder Jahan Parash

Name of the Team: Team KMAP

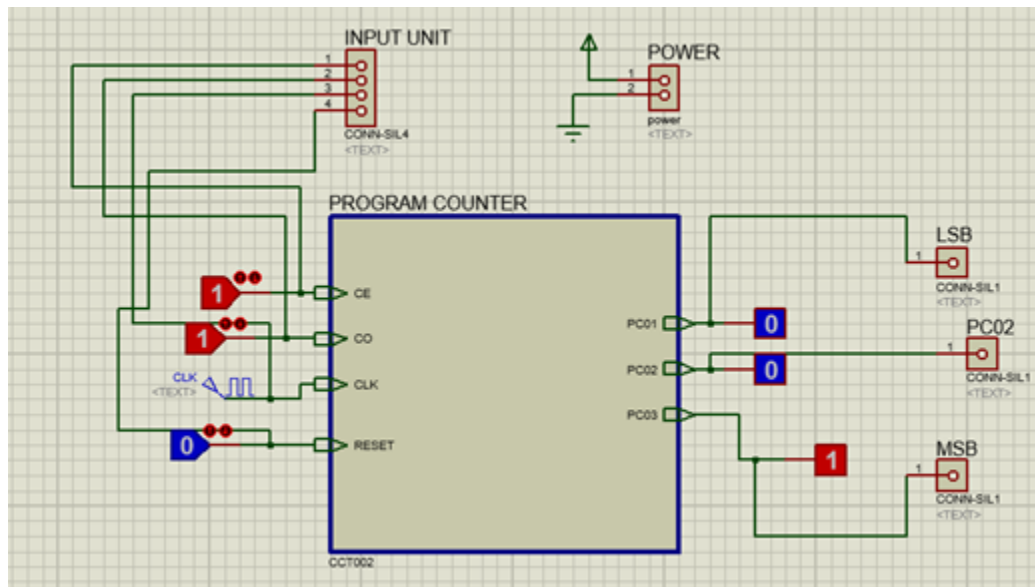
Team members:

- Maisha Fahmida Semonty 210021208
- Munshaat Atiq Nuzaba 210021210
- Mahzabin Ishrat 210021233
- Faria Ahmed Suprova 210021236
- Somia Alam 210021239
- Zaima Haque 210021259

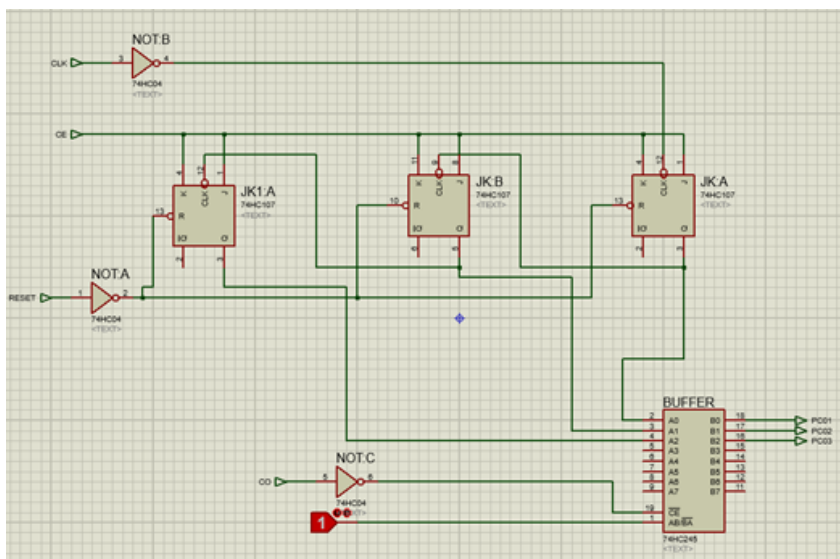
Program Counter:

Before connecting the program counter module to the bus, logic states were connected to the output wires of the program counter. Since the logicstates showed correct values, it was understood that the program counter was functioning accurately. Subsequently, the PC01 output from the module was connected to the least significant bit (LSB) of the bus, followed by the sequential connection of the other outputs. To ensure unidirectional data flow, a 74HC245 buffer IC was added between the output and the bus, ensuring that data could only travel from the output to the bus, preventing any data from returning to the program counter through the output lines.

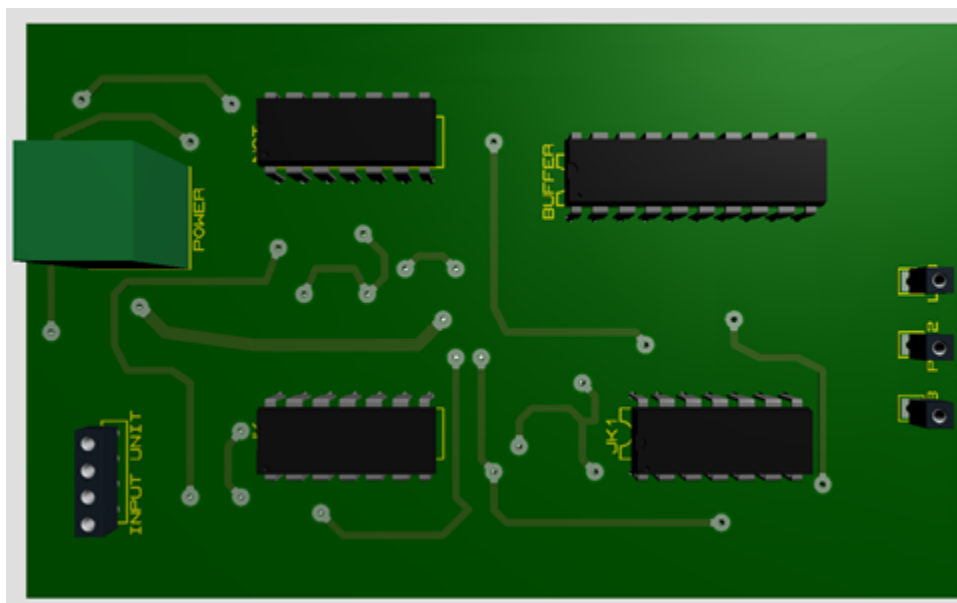
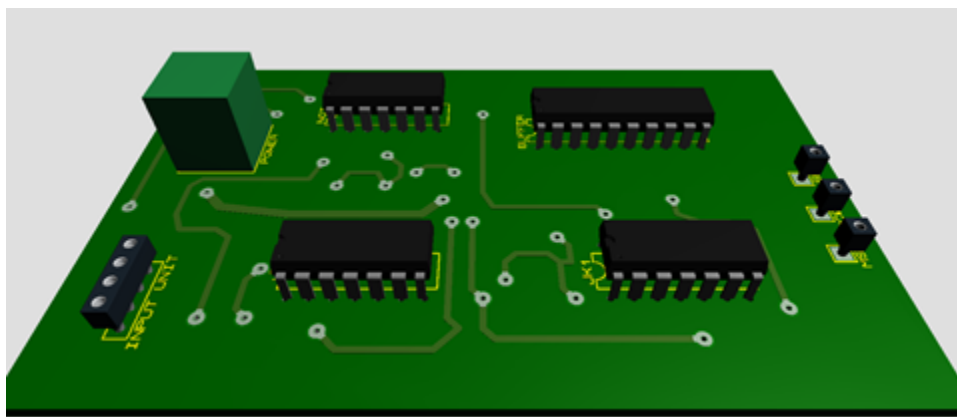
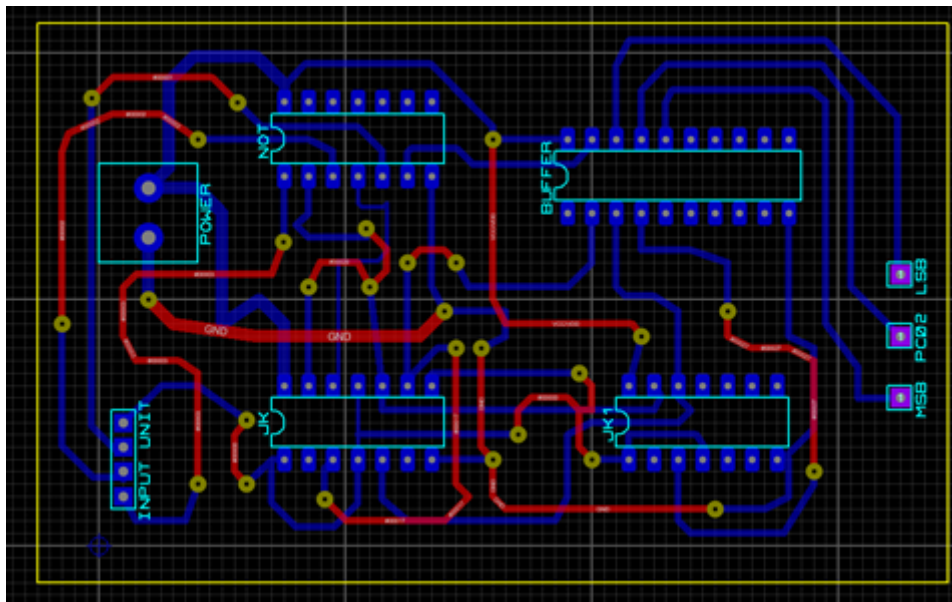
Parent Sheet:



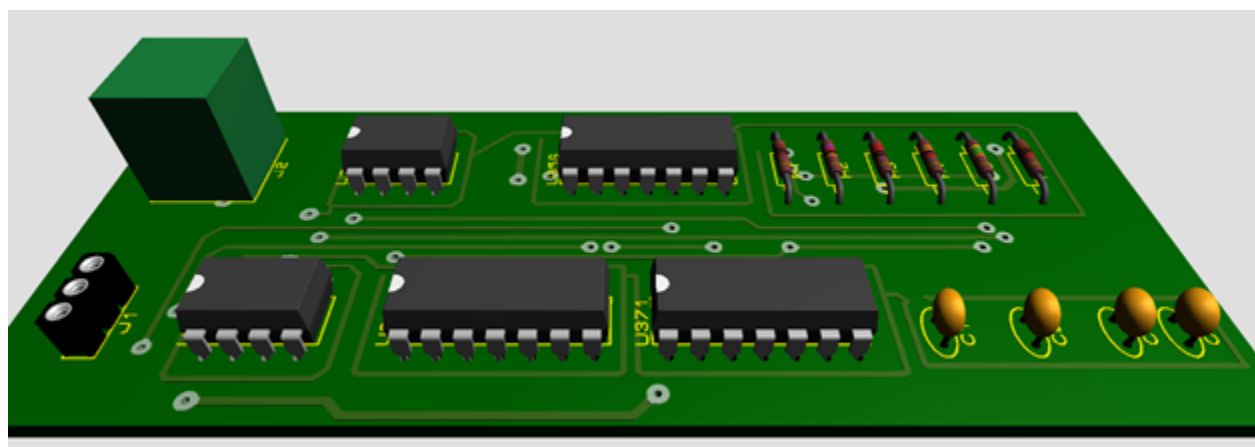
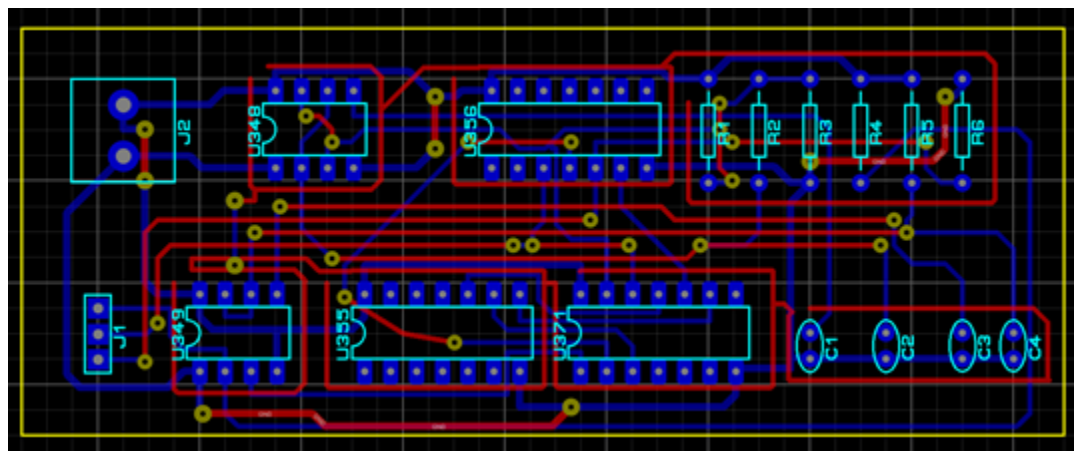
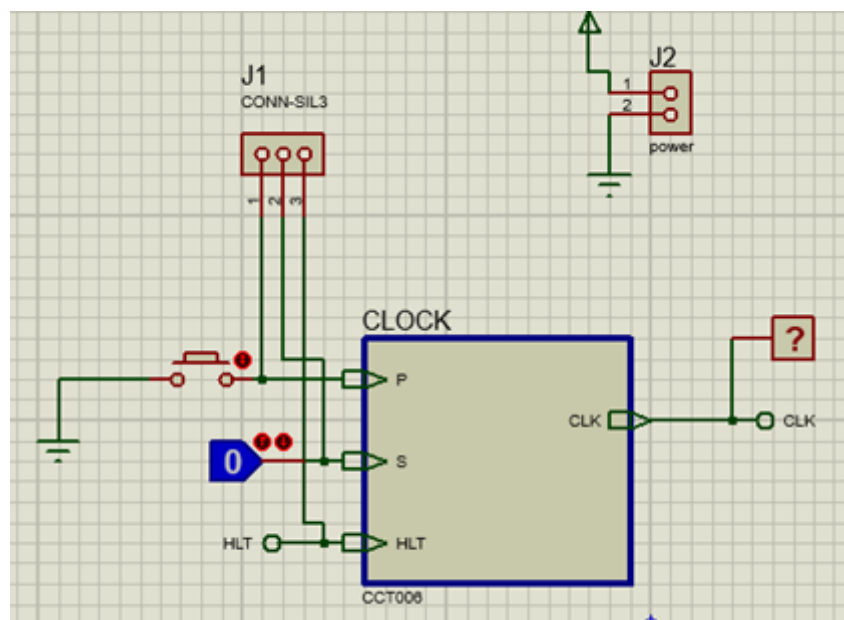
Child Sheet



PCB Layout

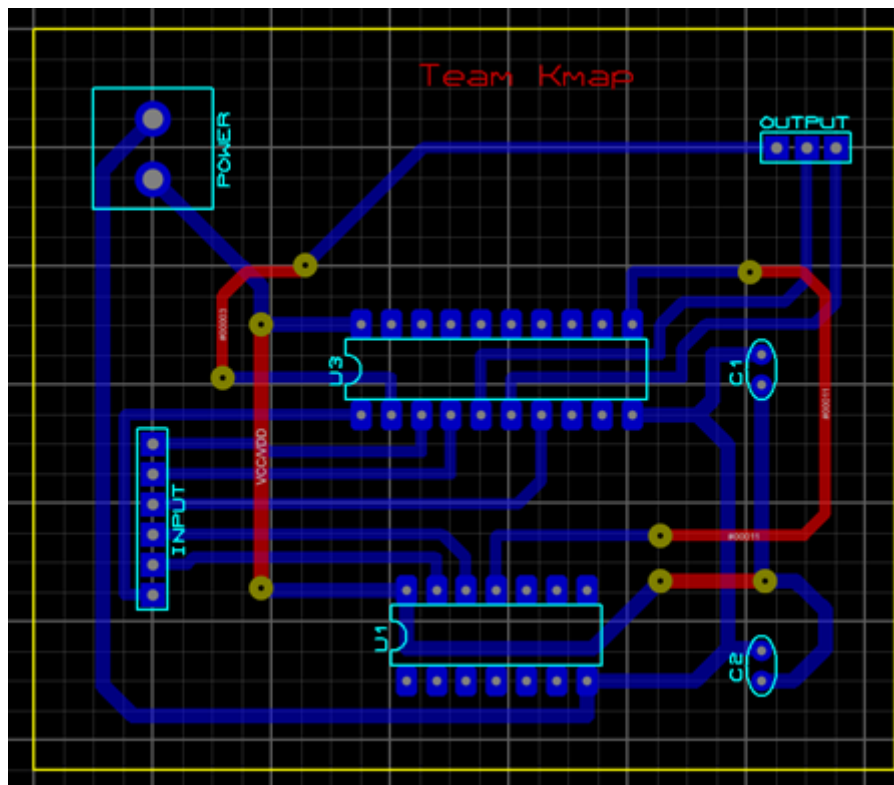
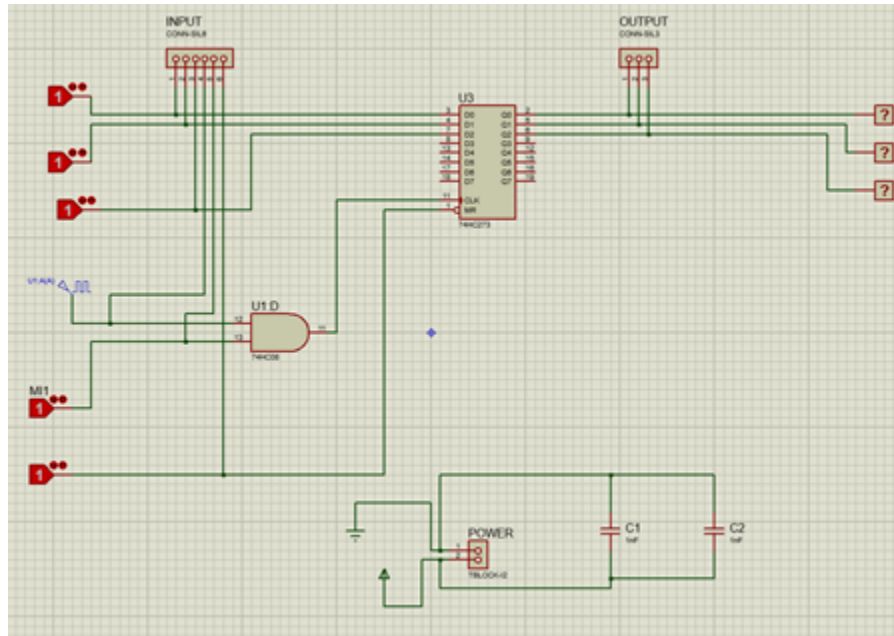


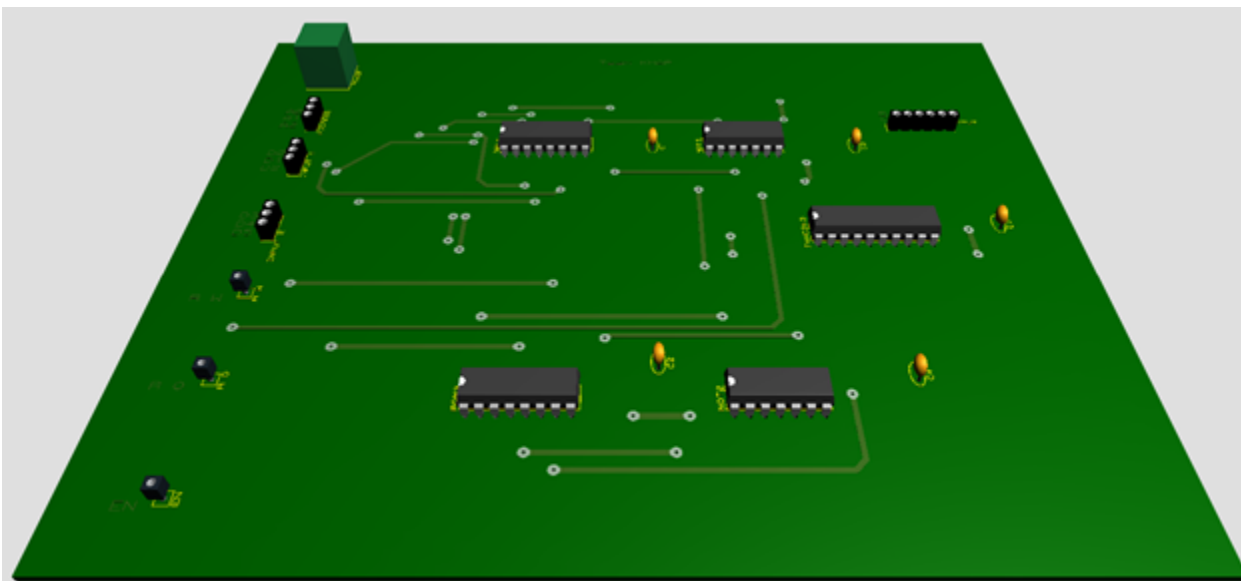
Clock Schematic:



Memory Address Register (MAR):

Firstly, the MAR module was created in the proteus file and manual connection was given to verify if it was working properly or not. After verifying it was connected to the bus. LSB of the MAR module was connected to LSB of the bus and connection was given sequentially.





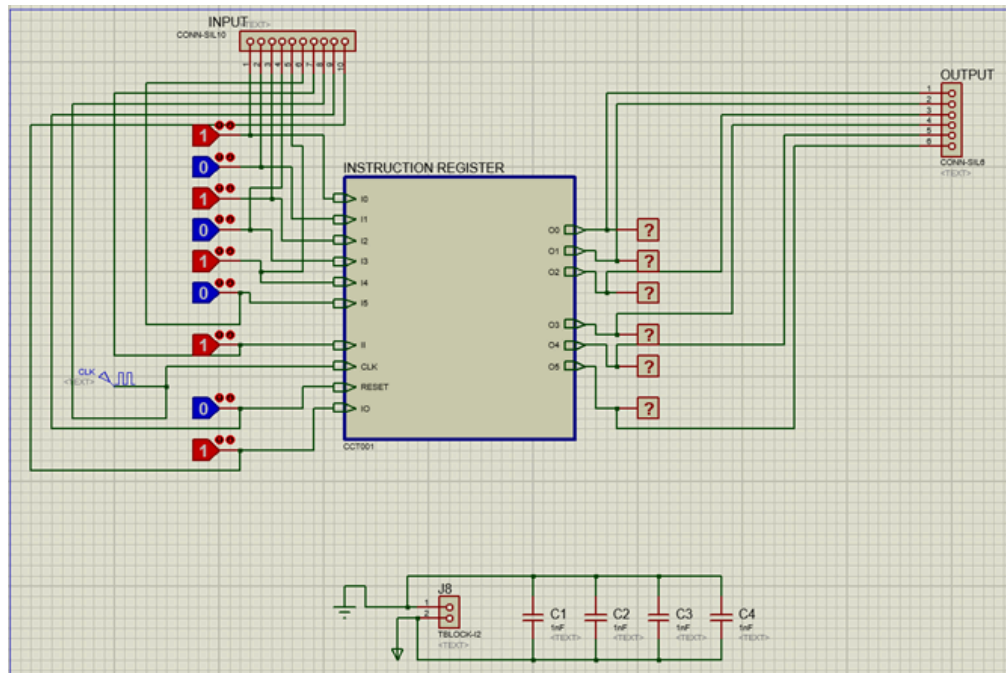
Instruction Register

Objective: The main purpose of this part of the project was to understand how an instruction register work in a computer; to construct a circuit of said register using Proteus and to design the PCB of the circuit.

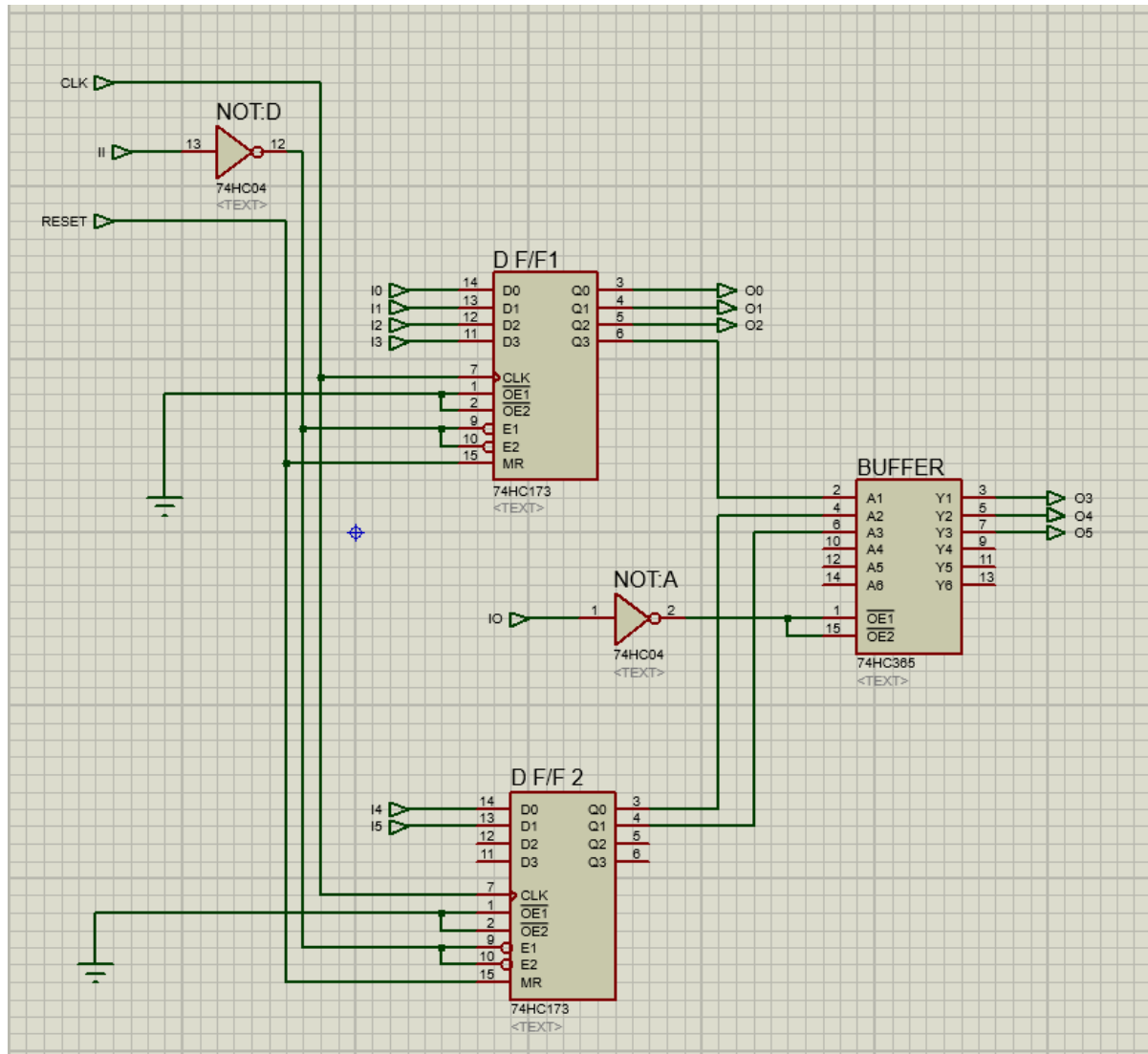
Description:

Pin	Name	Function
II	Instructio n Register In	Writes the current values of the bus into the instruction register.
IO	Instruction Register Out	Sends the stored data (only lower nibble) onto the bus. Output pins are disconnected when AO is low.

Parent Sheet:

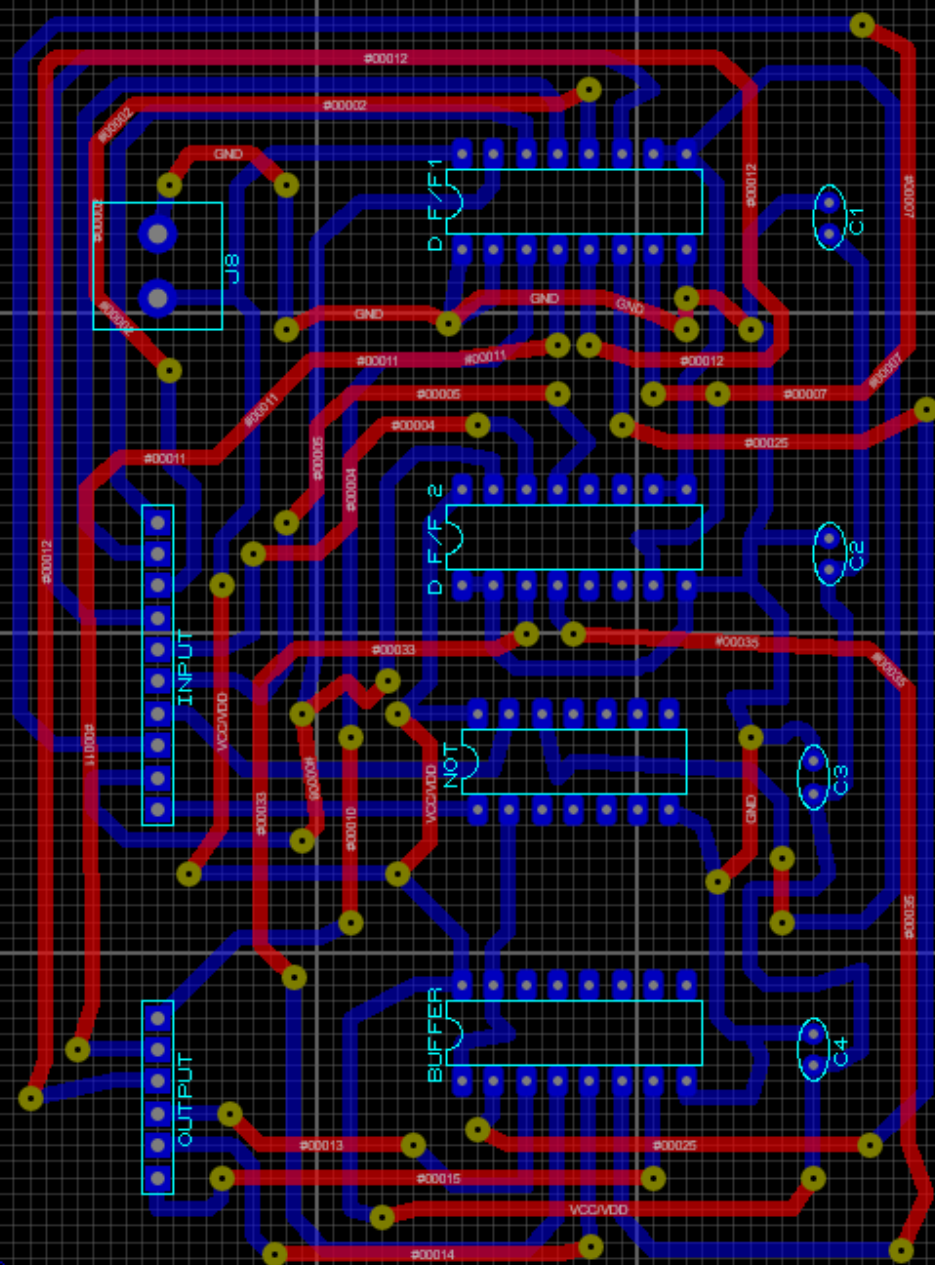


Child Sheet:

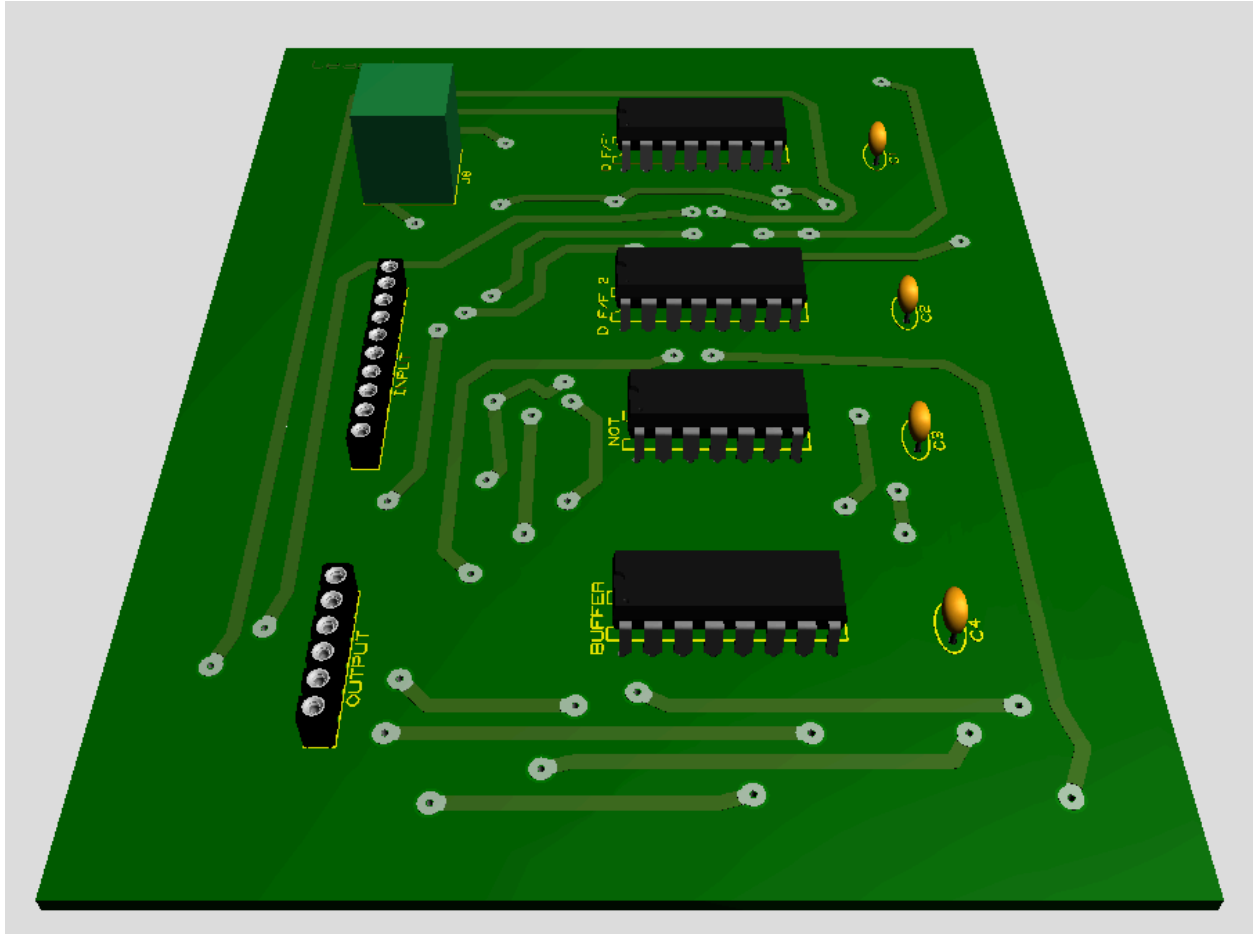


PCB LAYOUT:

team kmap



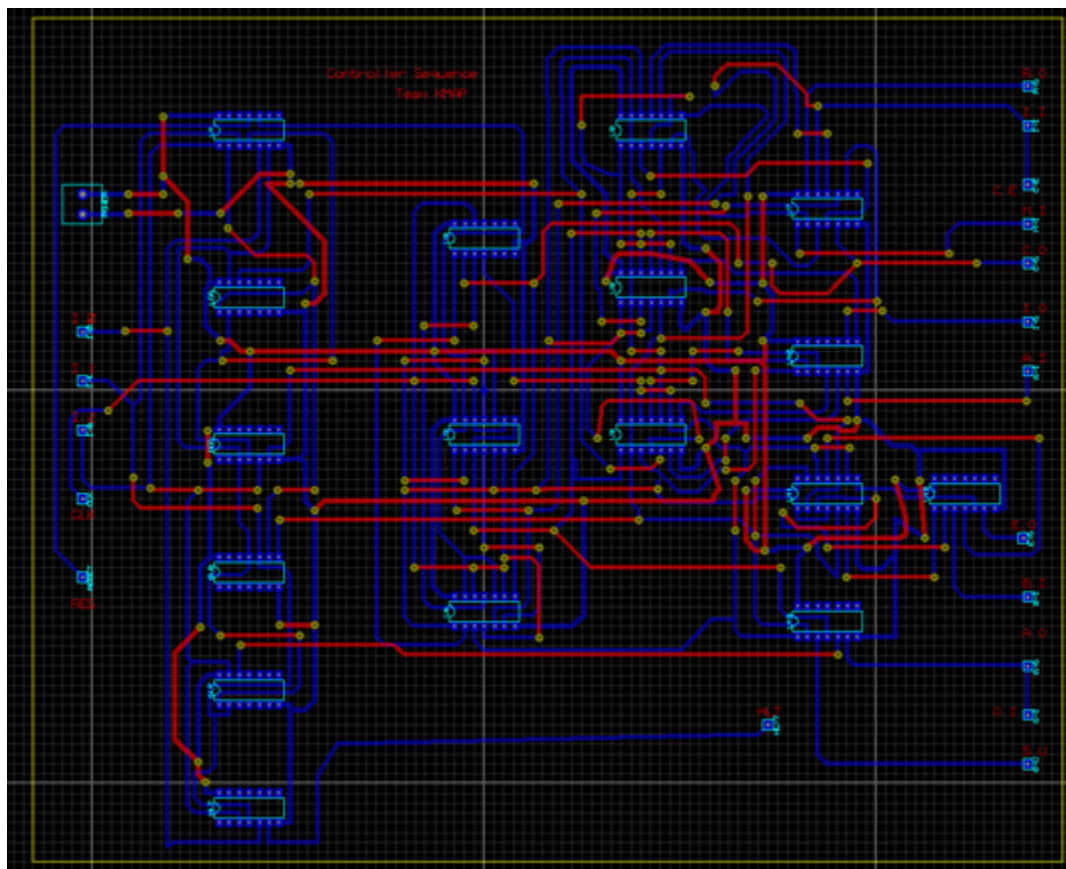
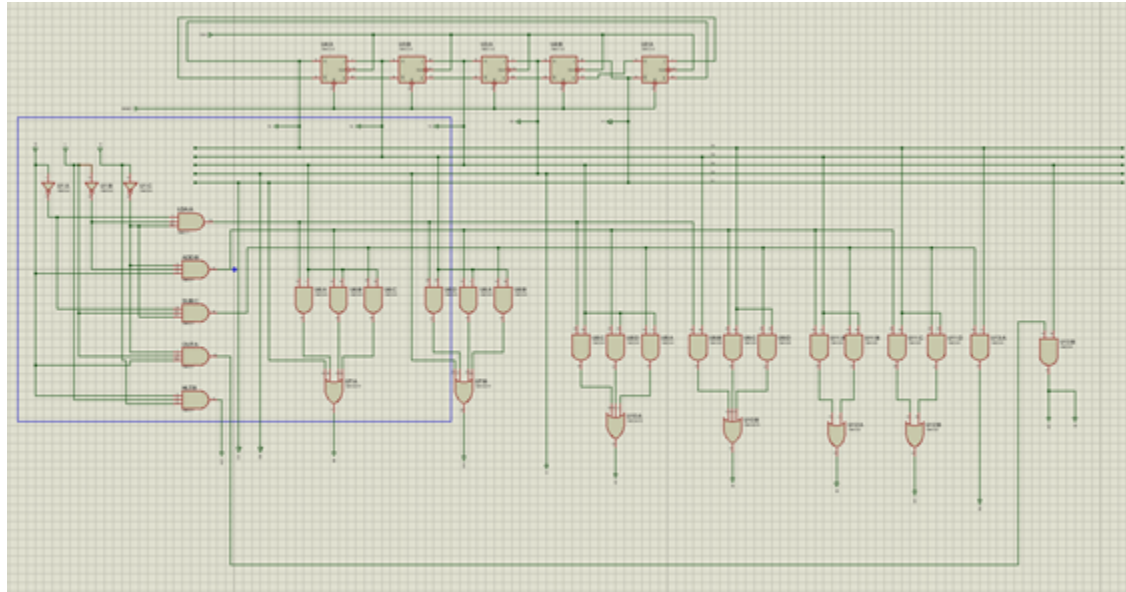
3D View:

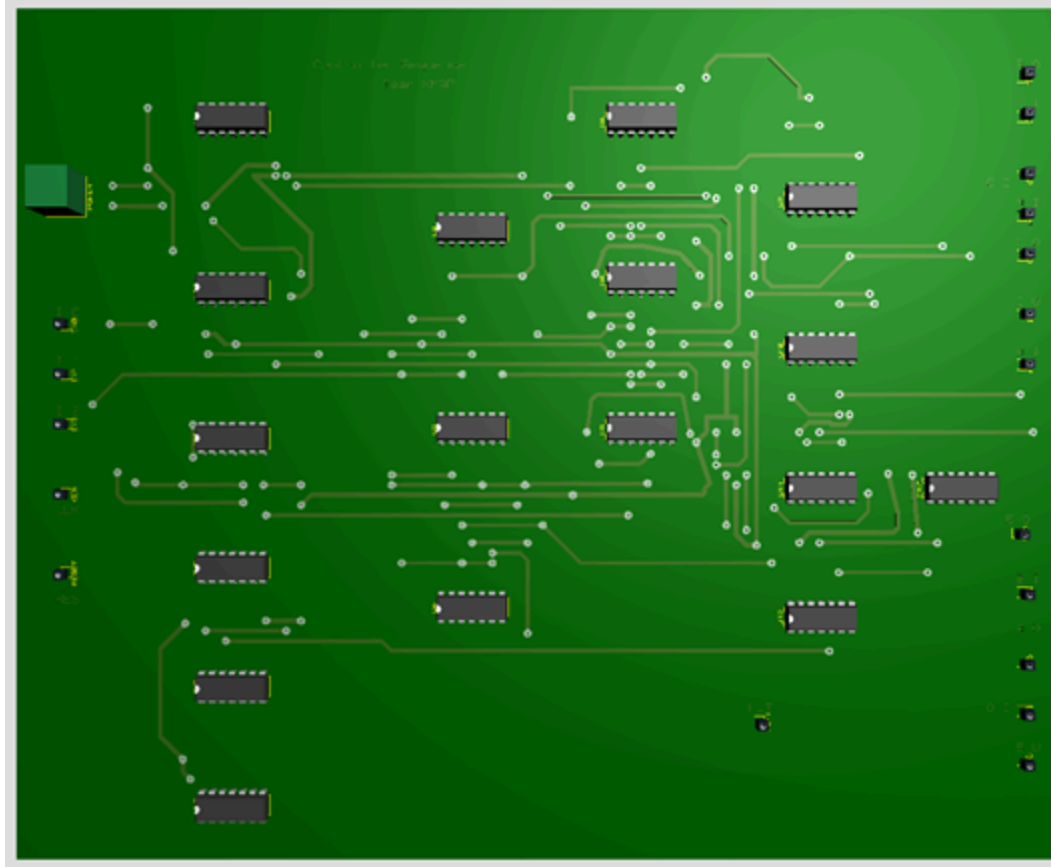


Controller Sequence:

After designing the controller sequencer module, it was connected to the instruction register. O3, O4, O5 went to the input I2, I1 and I0 respectively. Before connecting the wires coming from the instruction register, logicstate was connected to I0, I1 and I2 and opcodes were given manually. It was checked if the controller sequencer performed its operation properly. Logicprobe for T states were connected and after ensuring that each control bit gets activated for their specific T state, terminals were used removing the Logicprobes from the output and

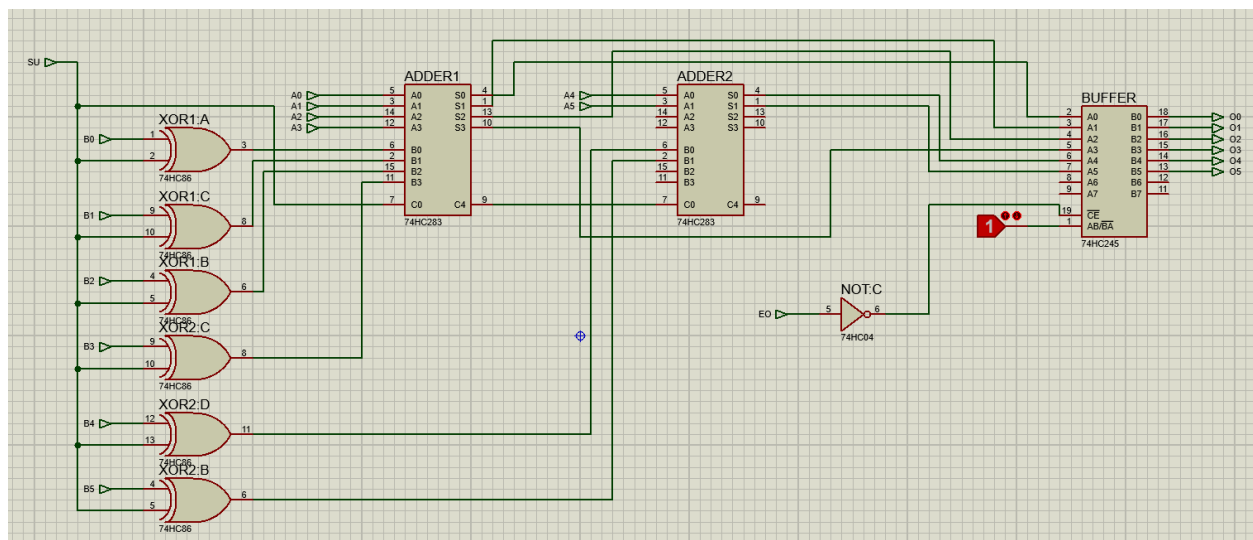
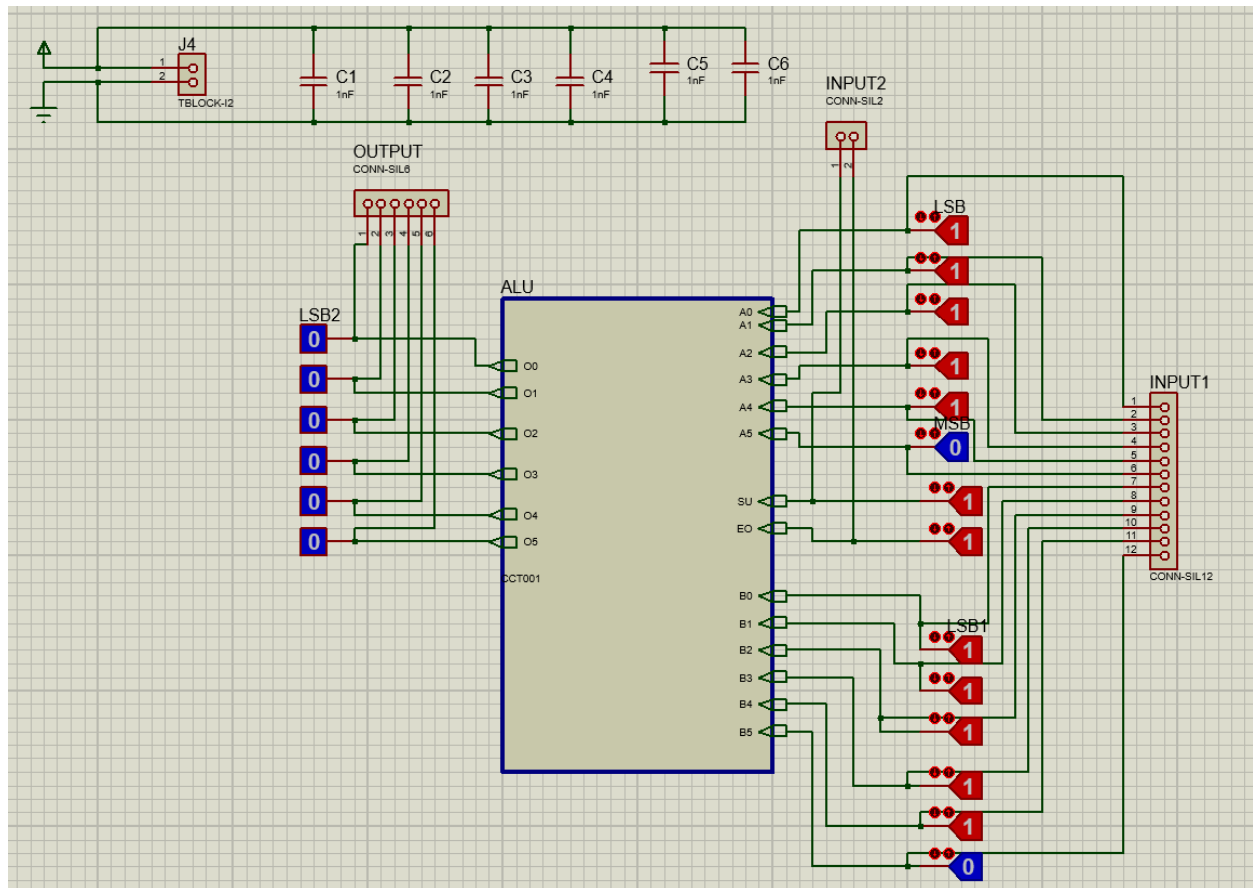
proper names were given for the each control bits. Connections from the instruction register were given after checking that the controller sequencer gets opcodes from the instruction register.

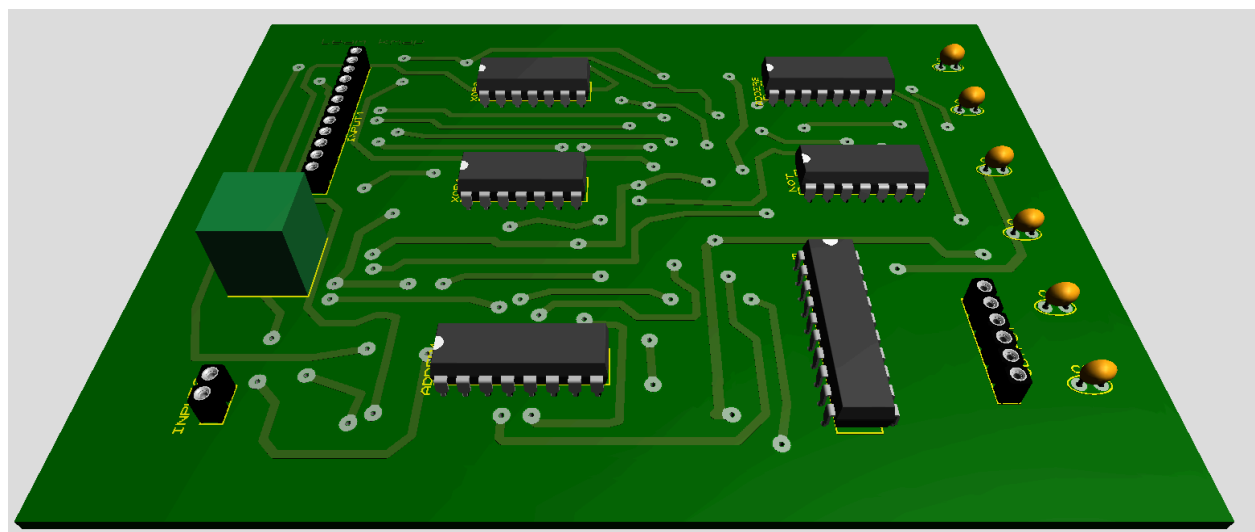
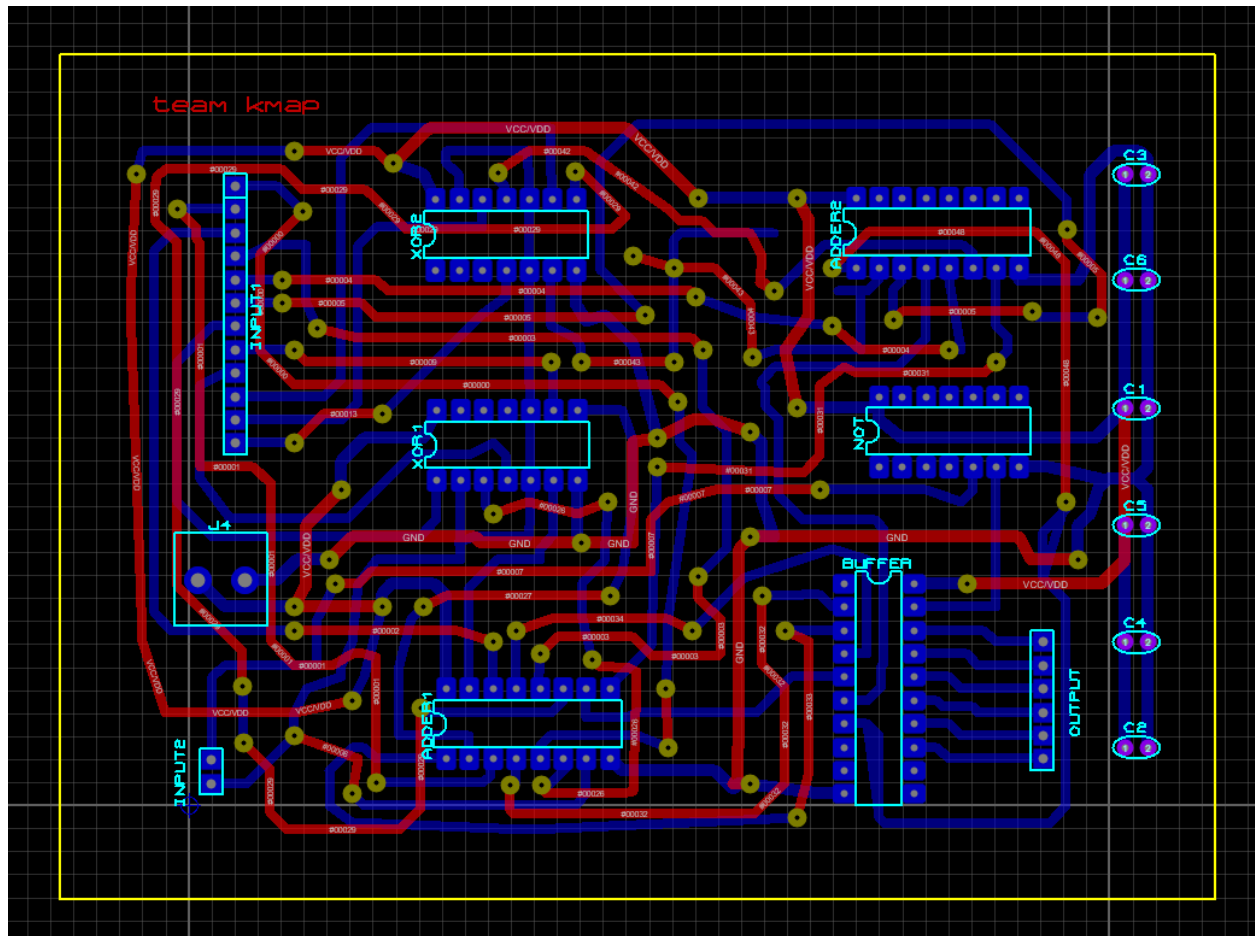




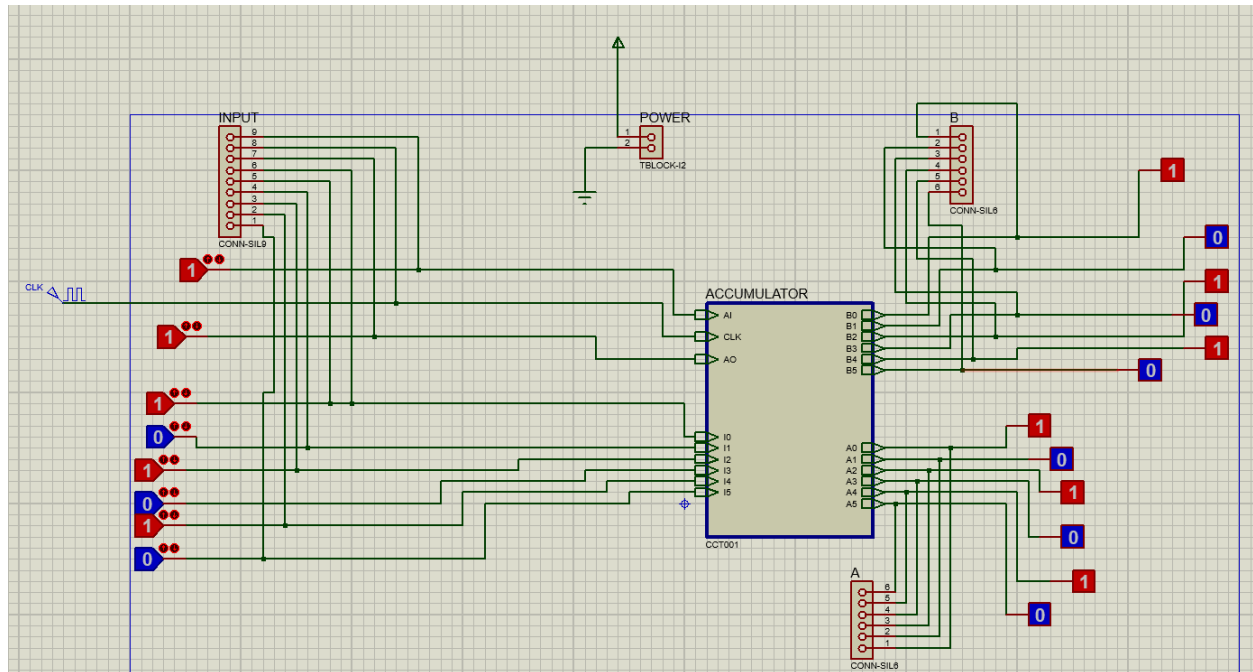
Arithmetic and Logic Unit (ALU):

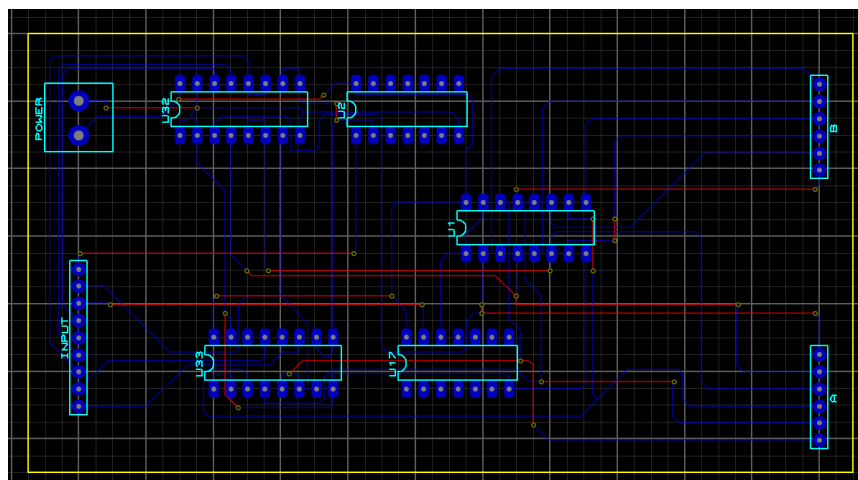
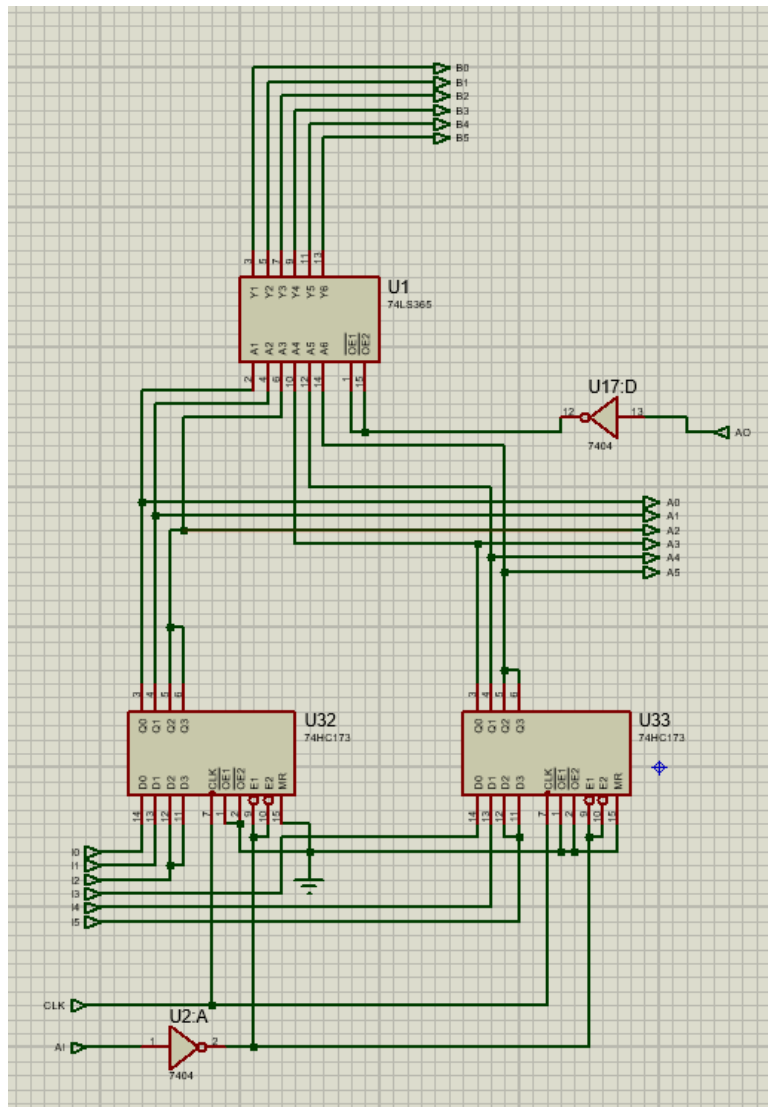
The submodule for ALU was built and then logicprobe and logicstate was connected to verify whether it works or not. inputs SU and EO comes from controller sequencer. At first the correct output was not showing properly. In cases where the output should be 000000 , ALU was showing 111000. To solve this problem the buffer ic was changed from 74HC365 to 74HC245. After this change there was no issue with the unit. Input A0 to A5 was connected with A0 to A5 of accumulator. Similarly input B0 to B5 is directly connected with the output of B register. Then the output was connected with the bus.O0 is LSB and O5 MSB. To check whether data is properly going to the bus we added a switch between the output and bus. Logicprobes were connected either side of the switch to verify this.

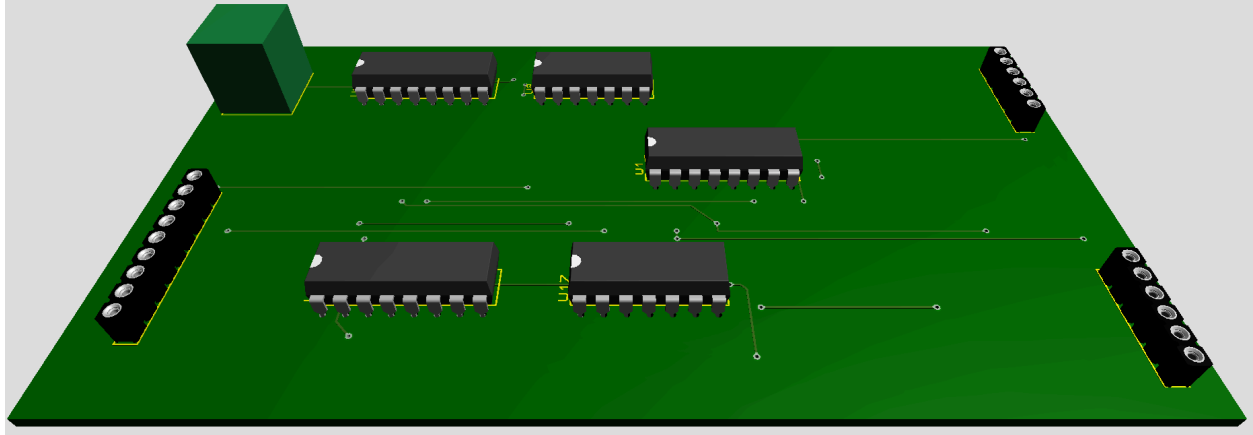




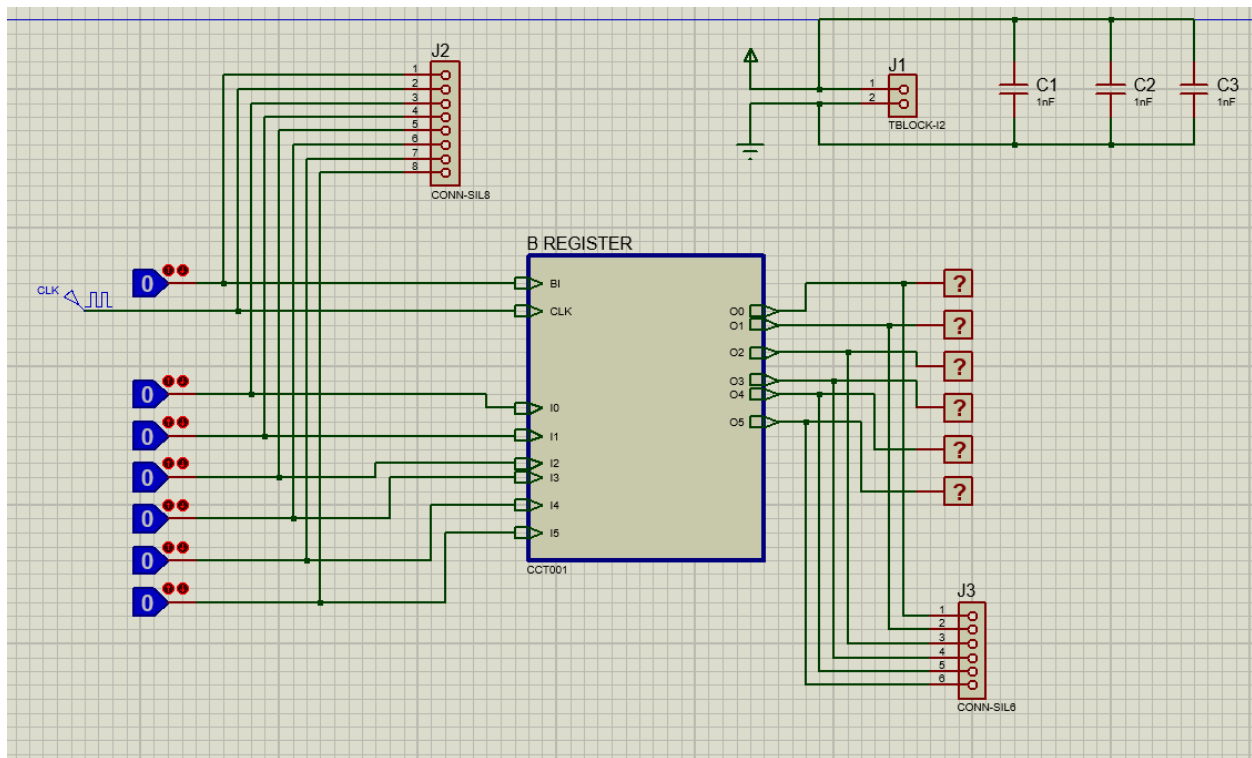
Accumulator: After creating the accumulator module, manual connections were given to check whether it is functioning properly or not. The 74HC04 ics which were connected in the child sheet to the E1, E2 pin of the 74HC173 ics and OE1' and OE2' pins of 74HC365 IC through AI and AO, these ICs were shifted to the parent sheet. After ensuring the accumulator was functioning properly, wires were connected to the accurate position from the bus via some switches. Here switches were used to check the unidirectional flow of data.

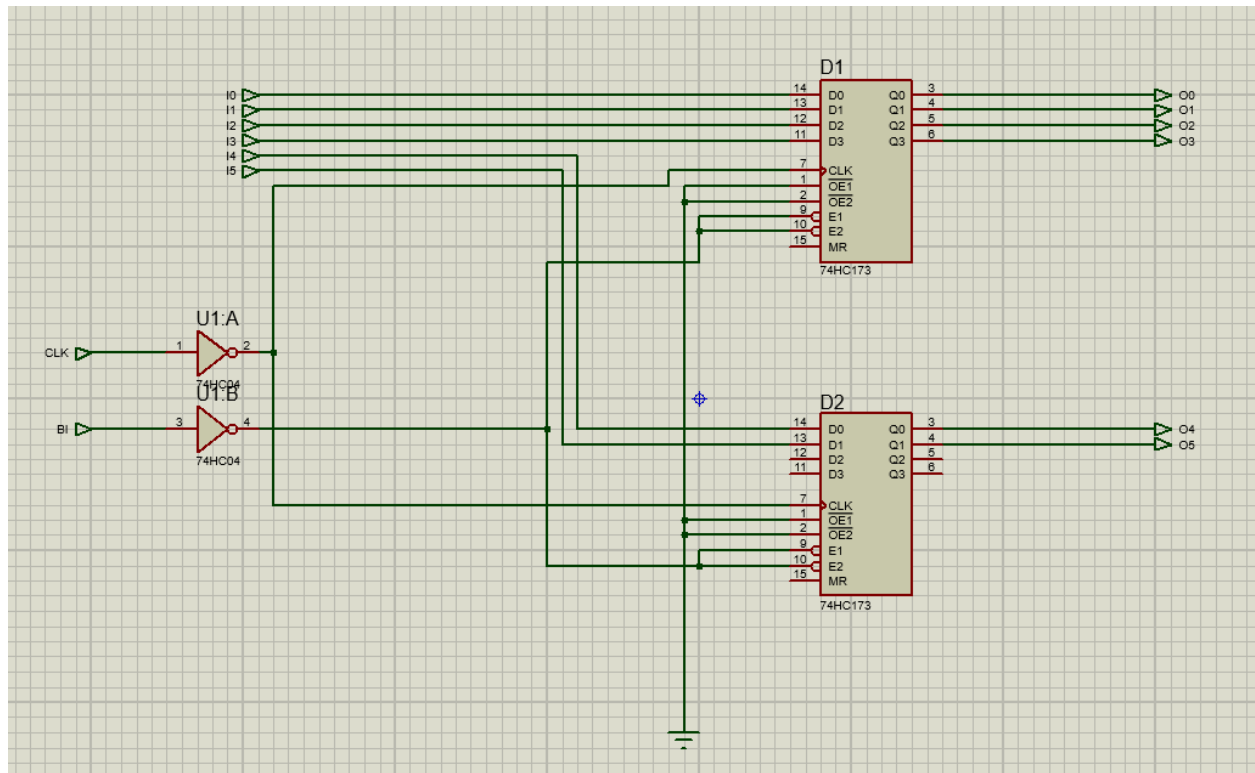


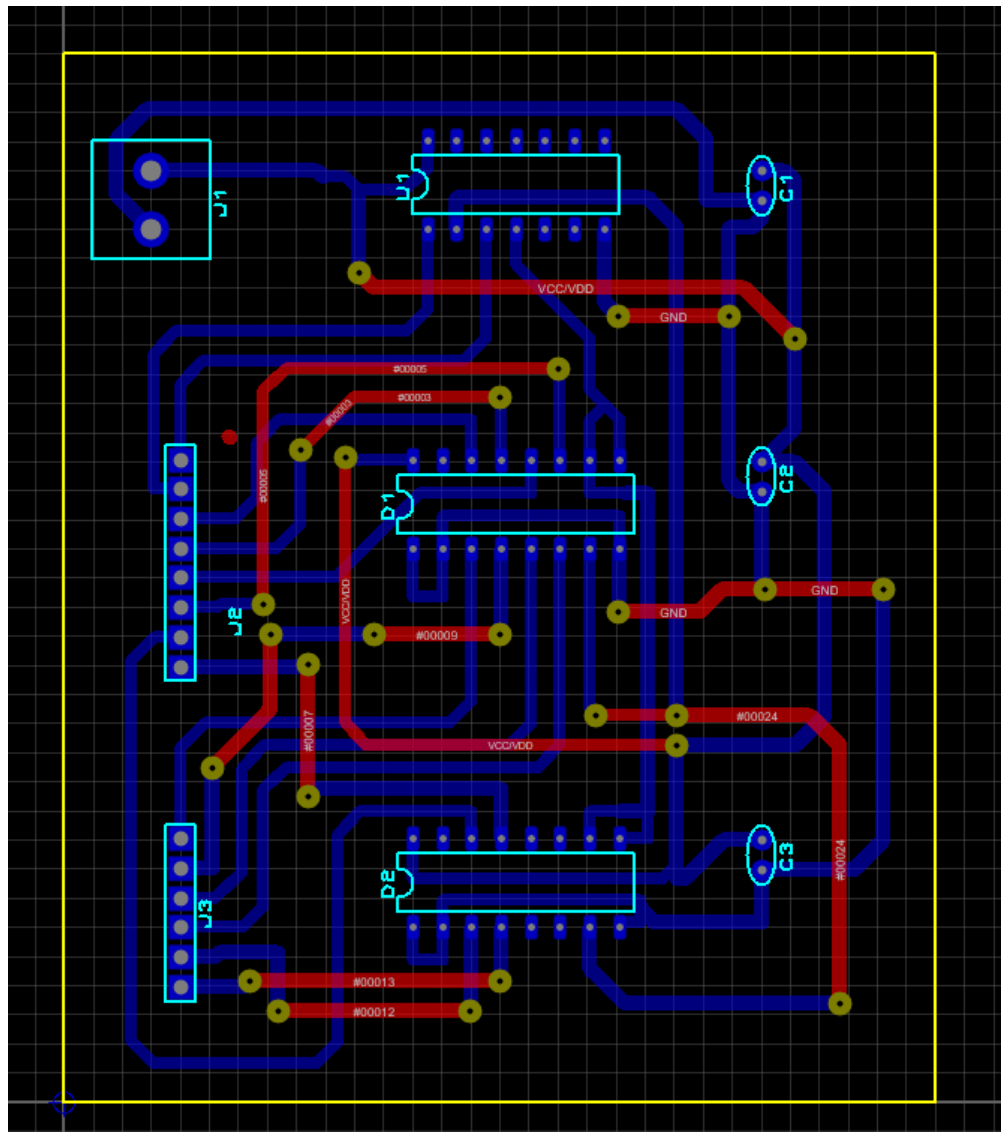


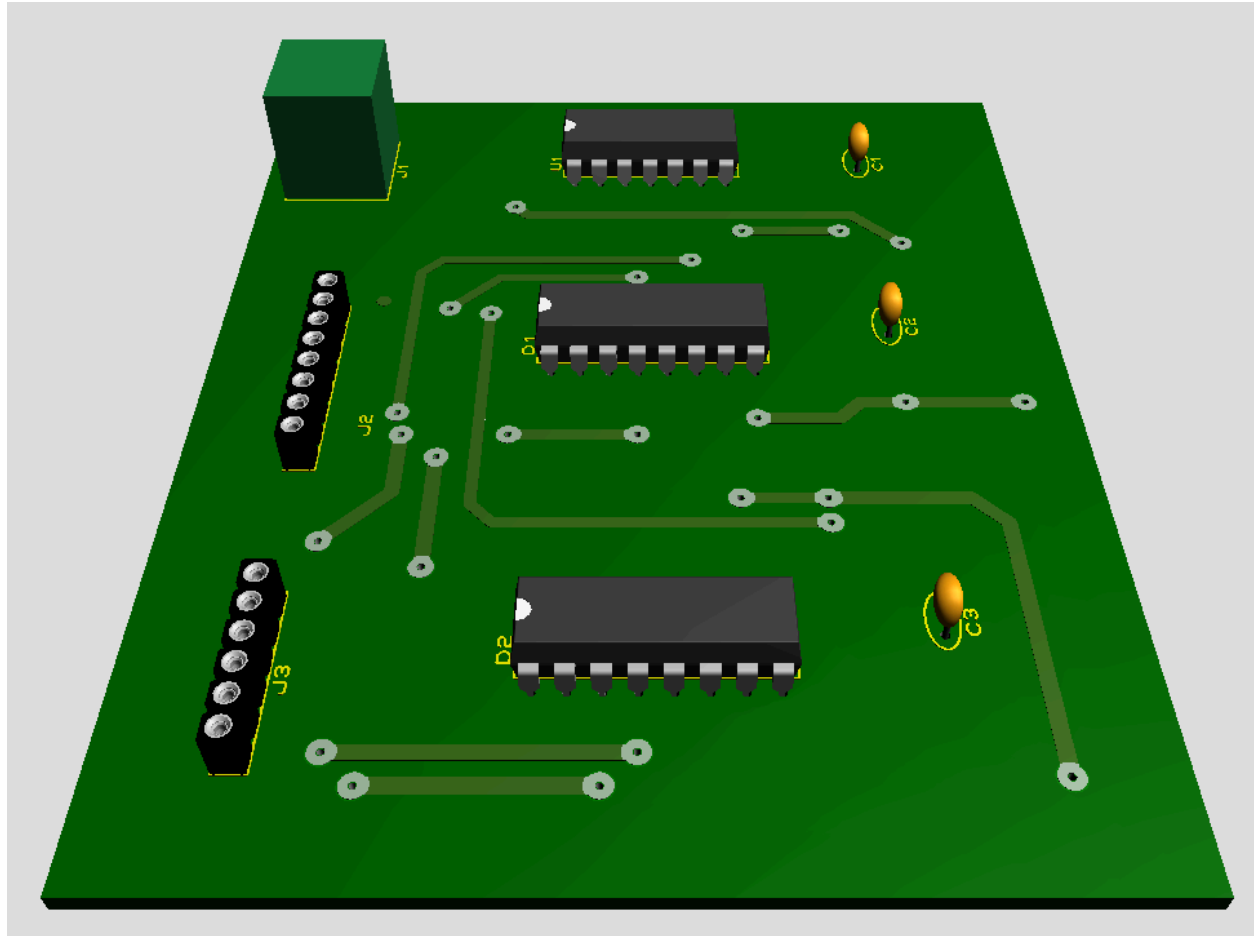


B Register: After creating the module in the proteus file, manual connections were given to check whether the register functions properly or not. Also it was checked whether the control bit BI is getting signal from the controller sequencer or not. After ensuring everything, connections were given to the accumulator and bus removing the manual connections. Here bus sends signal to the B register and B register sends the signals to the ALU.

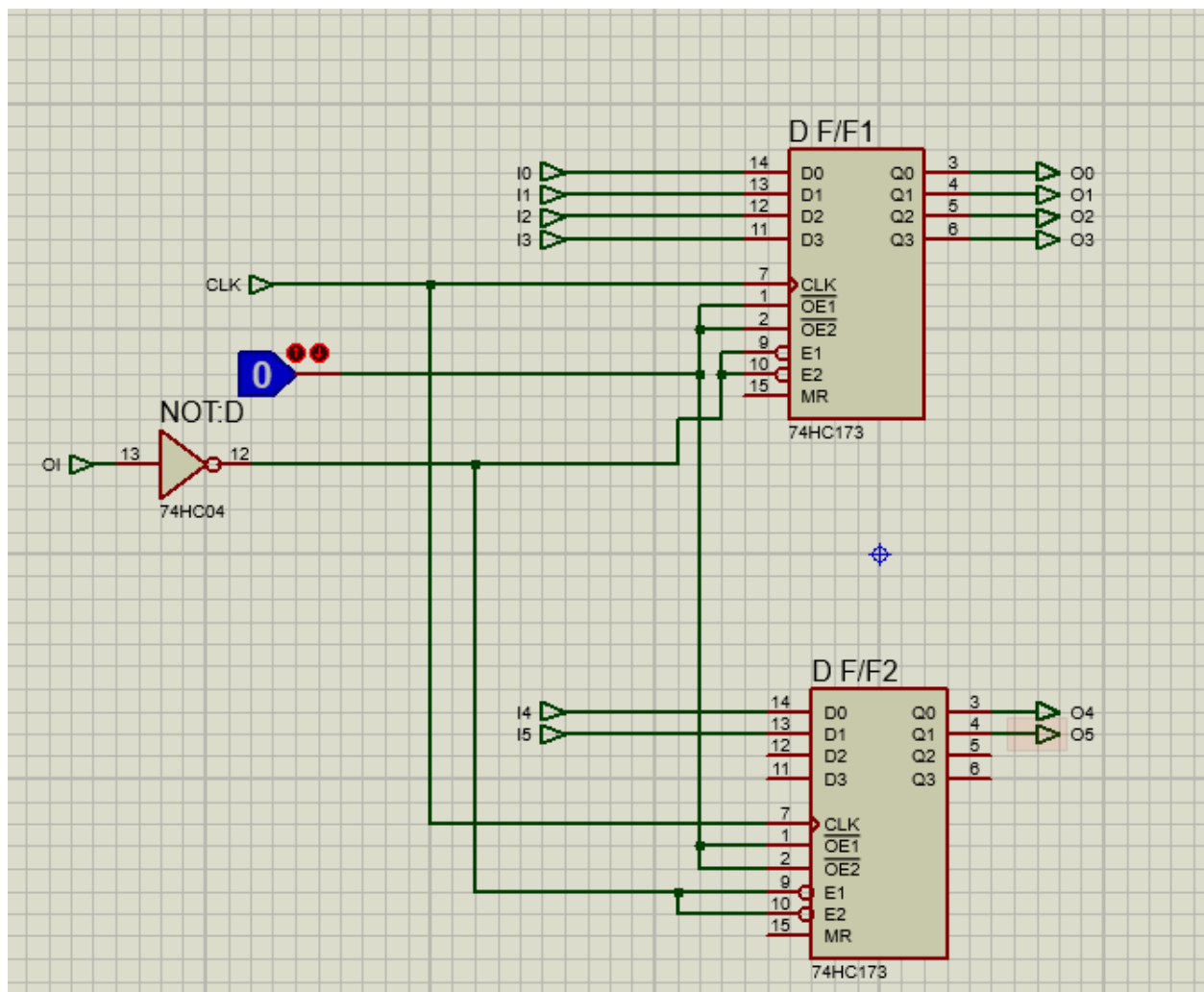
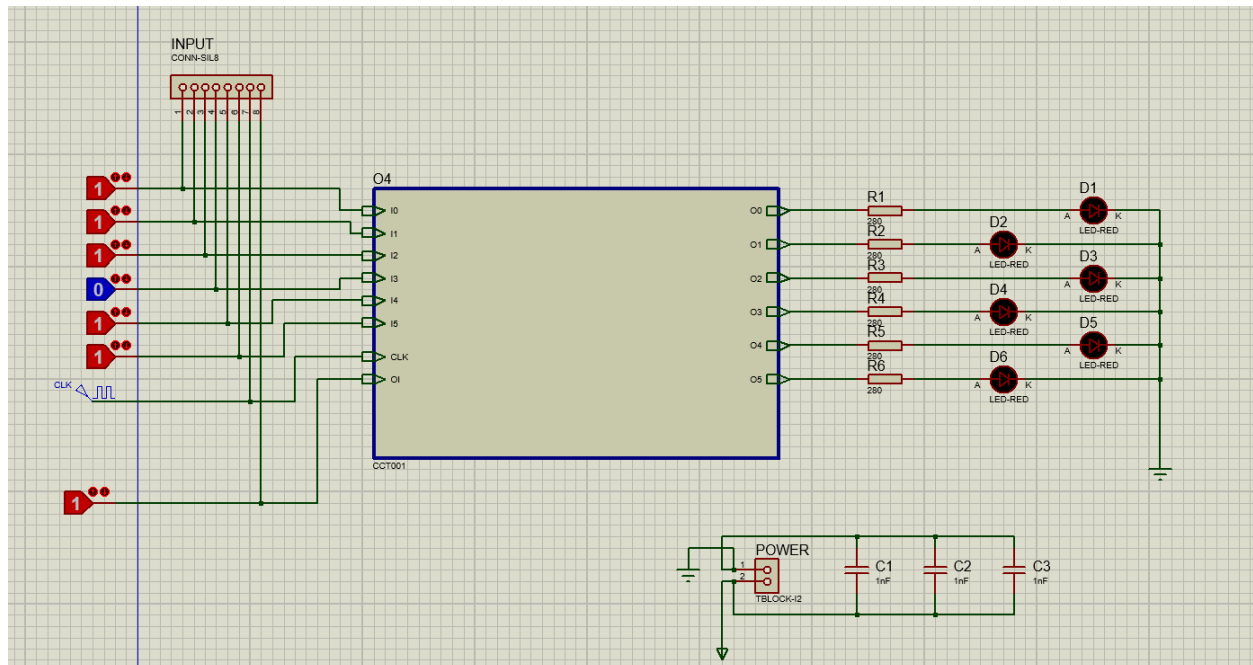


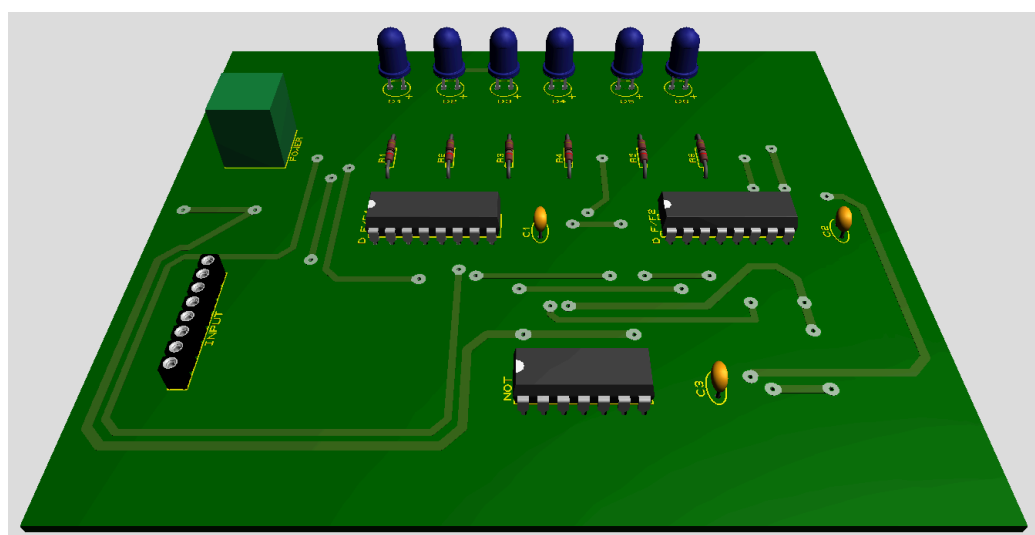
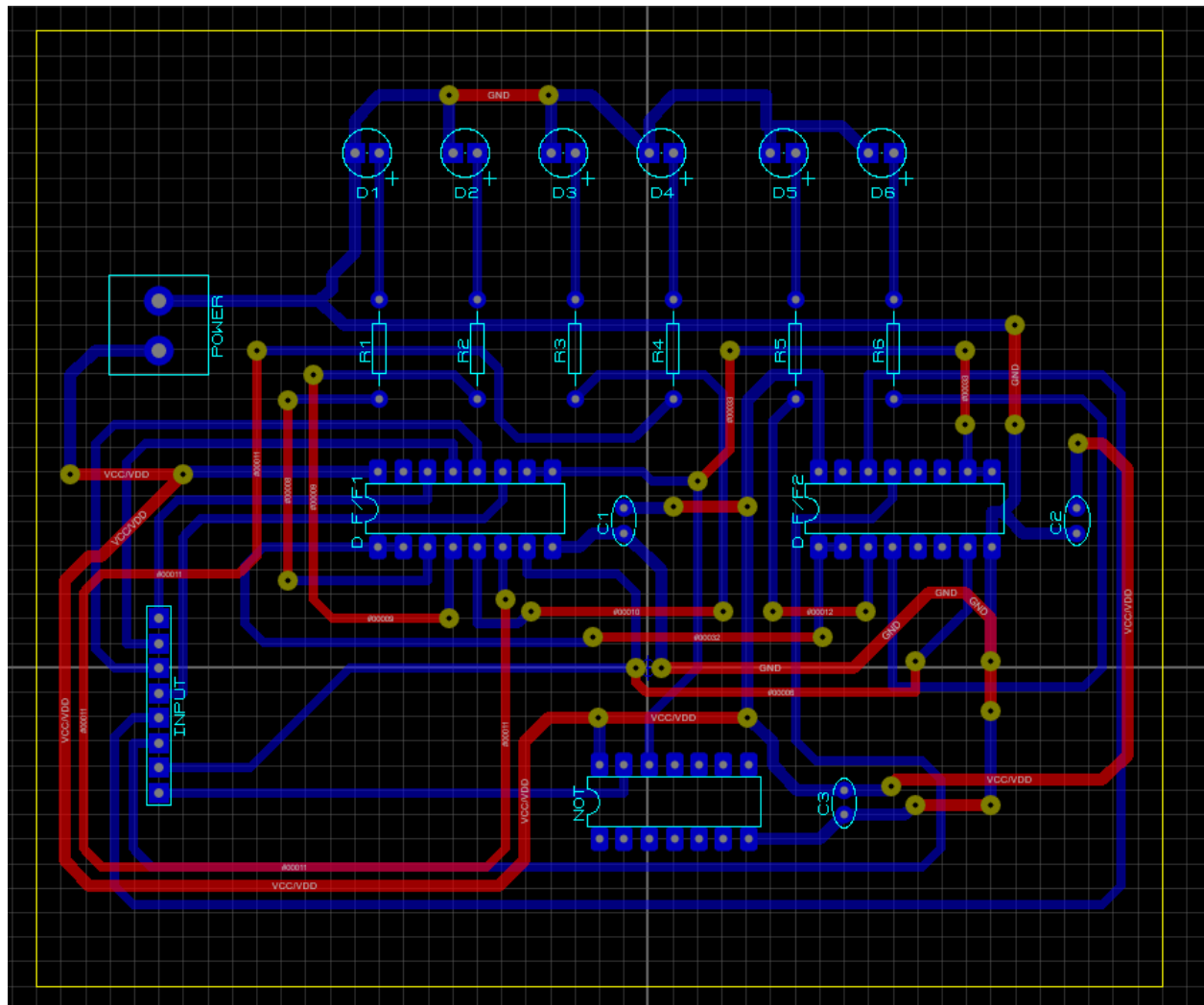






Output register: First the submodule for the output register was built and then with the help of logicprobes and logicstates it was checked whether it worked properly or not. After seeing that it works the output register was connected to the bus. LSB of the bus was connected to I0 and other connections were given sequentially. CLK input of the register comes from the main clock of SAP and OI input comes from controller sequencer. To verify if the SAP-1 module is working properly or not, first clock was turned off and all modules were reseted. RAM was selected to Prog mode and inputs were given on the specific addresses. After giving the input datas, Run mode was selected and the clock was again turned on. After some time output unit showed results for addition and subtraction.





Overall:

